Relations, partial orders, equivalence, and machine minimization

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Basic properties of relations

For $R \subseteq A^2$, reflexivity means $\forall a \in A, (a, a) \in R$; symmetry means $\forall a, b, aRb \Rightarrow bRa$; and transitivity means $\forall a, b, c, (aRb \land bRc) \Rightarrow aRc$. Irreflexivity forbids every loop and is not the logical negation of reflexivity. Antisymmetry requires $(aRb \land bRa) \Rightarrow a = b$, and differs from asymmetry. For example, $\leq$ is reflexive and transitive without being symmetric.

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Asymmetry differs from antisymmetry

Asymmetry requires $aRb \Rightarrow \neg bRa$, including $a = b$, so it forbids loops. It is equivalent to antisymmetry plus irreflexivity. The first forbids opposite edges between distinct points; the second forbids the same-point case.

Equality $\Delta_A = \{(a, a) : a \in A\}$ is both symmetric and antisymmetric. On a nonempty set, $\leq$ is antisymmetric but not asymmetric. The empty relation on nonempty A is symmetric, antisymmetric, transitive, and irreflexive, but not reflexive.

Transitivity allows repeated variables: aRb and bRa force aRa. Checking only three distinct points misses this. On the empty set the unique relation is both reflexive and irreflexive, since neither universal assertion has a counterexample.

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Counting finite relations

For $|A| = n$, each ordered pair is optional, giving 2^{n^2} relations.

Requirement	Independent choices	Count
Reflexive	Fixed diagonal; other pairs free	$2^{n(n-1)}$
Symmetric	Free diagonal; opposite directions chosen together	$2^{n+\binom{n}{2}}$
Reflexive and symmetric	Fixed diagonal; free unordered pairs	$2^{\binom{n}{2}}$
Antisymmetric	Free diagonal; neither, left, or right per distinct pair	$2^n 3^{\binom{n}{2}}$
Asymmetric	Same choices without diagonal	$3^{\binom{n}{2}}$
Symmetric and antisymmetric	Only diagonal pairs optional	2^n

Adding reflexivity in the last row leaves equality alone. Transitivity couples choices of different pairs and prevents independent multiplication.

Composition direction and matrices

Use the usual function direction. For $R \subseteq A \times B, S \subseteq B \times C$,

$$(a, c) \in S \circ R \iff \exists b\ (aRb \wedge bSc).$$

Apply R first, then S. Some references use left-to-right notation; the pair definition determines the convention.

Both associative expressions mean $\exists b, c\ (aRb \wedge bSc \wedge cTd)$, proving $T \circ (S \circ R) = (T \circ S) \circ R$. Reversing the route gives $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.

With source rows and destination columns, $M_R \odot M_S = M_{S \circ R}$. Integer multiplication counts two-step walks; Boolean multiplication asks whether at least one exists. Entrywise, reflexivity means $I \leq M$, symmetry $M = M^T$, transitivity $M \odot M \leq M$, and antisymmetry $M \wedge M^T \leq I$.

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Zero-one matrices

Define $M_{ij} = 1$ iff $(a_i, a_j) \in R$. Reflexivity means an all-one diagonal; symmetry means $M = M^T$. Boolean composition uses

$$(A \odot B)_{ij} = \bigvee_k (A_{ik} \wedge B_{kj}).$$

The reverse relation has the transposed matrix. Transitive closure is the union of Boolean powers. On n vertices, positive reachability uses $R \cup \cdots \cup R^n$: distinct endpoints need at most n−1 edges in a simple path, but a nonempty return cycle may need n. Reflexive-transitive closure uses $I \cup R \cup \cdots \cup R^{n-1}$. Boolean addition is OR and multiplication AND, unlike integer multiplication.

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Powers of a three-vertex digraph

Take $a \to a, a \to b, b \to c, c \to a$, indexed a,b,c:

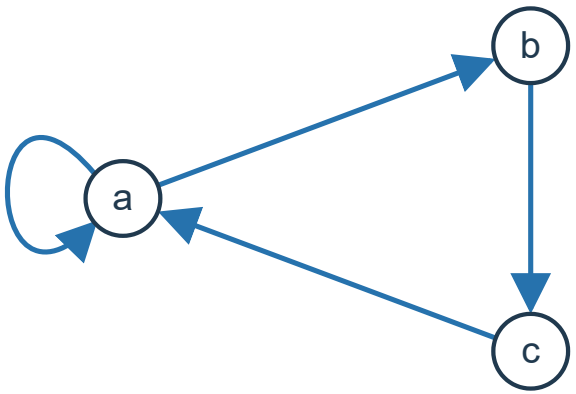
$$M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad M^{\odot 2} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \quad M^{\odot 3} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

The square's bb entry is zero because c does not return directly to b. The cube's bb entry is one via $b \to c \to a \to b$; c also first returns in three steps.

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Powers of a three-vertex digraph (continued)

Three-point relation: edges and length-2,3 walk endpoints



Original M: 100 OR 010 = row a, 110

M^2 rows : 111 / 100 / 110

M^3 rows : 111 / 110 / 111

b → c → a → b: first positive return

$R^+ = R \cup R^2 \cup R^3$: all 9 pairs

OR-ing the three powers gives all ones. The square alone is not the closure. Relations are sets of pairs, allowing loops and opposite edges but no parallel copies of a pair.

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Closure existence and minimality

Let R^k contain endpoints of exactly length-k walks and $R^0 = \Delta_A$. Define

$$R^+ = \bigcup_{k \geq 1} R^k, \qquad R^* = \Delta_A \cup R^+.$$

Concatenating positive walks proves R⁺ transitive and containing R. If transitive T contains R, induction on walk length gives every $R^k \subseteq T$, so $R^+ \subseteq T$. This proves minimality; R* similarly is the smallest reflexive-transitive superrelation.

On $n \geq 1$ vertices, delete internal cycles between distinct endpoints to leave at most n−1 edges. A shortest positive return is a cycle of at most n edges. Hence

$$R^+ = R \cup \cdots \cup R^n, \qquad R^* = \Delta_A \cup R \cup \cdots \cup R^{n-1}.$$

The diagonal of a directed three-cycle shows why the first union cannot always stop at n−1.

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Warshall's invariant and in-place correctness

Start at $W^{(0)} = M_R$ and number vertices 1,…,n:

$$W_{ij}^{(k)} = W_{ij}^{(k-1)} \vee (W_{ik}^{(k-1)} \wedge W_{kj}^{(k-1)}).$$

An entry is one iff a positive i-to-j walk has internal vertices among 1,…,k. A shortest eligible walk either avoids internal k, or splits at k into two segments internally using earlier vertices. Conversely concatenate such segments. If an endpoint is k, a shortest walk needs no internal revisit, so the first term already retains it.

```
W := copy(M)
for k = 1..n:
    for i = 1..n:
        for j = 1..n:
            W[i,j] := W[i,j] OR (W[i,k] AND W[k,j])
```

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At fixed k, its row and column update as $x \vee (x \wedge y) = x$. Needed entries therefore stay unchanged, justifying in-place updates. Time is $O(n^3)$, matrix space $O(n^2)$. For R*, initialize the diagonal to one.

Every round on a four-vertex chain

Use edges a→b,b→c,c→d and indices a,b,c,d.

Round	Intermediate	New pairs	Four matrix rows
0	None	Original	0100 / 0010 / 0001 / 0000
1	a	None	0100 / 0010 / 0001 / 0000
2	b	(a,c)	0110 / 0010 / 0001 / 0000
3	c	(a,d),(b,d)	0111 / 0011 / 0001 / 0000
4	d	None	0111 / 0011 / 0001 / 0000

Round three uses the preceding round's a→c to obtain a→d, rather than only original edges. The acyclic graph's R^{+} diagonal is zero; R* adds loops.

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Every round on a four-vertex chain (continued)

Reflexive closure is $R \cup \Delta$; symmetric closure is $R \cup R^{-1}$. Equivalence closure is $(R \cup R^{-1})^*$. Reversing walks gives symmetry. Every equivalence superrelation contains inverses, identities, and concatenations, so contains this expression. Equivalently, ignore directions, find components, and include every within-component pair.

The intersection of a **nonempty** family of relations on a fixed set A preserves reflexivity, symmetry, antisymmetry, and transitivity separately. For antisymmetry, if both aRb and bRa lie in the intersection, choose one relation in the family and use its antisymmetry to conclude a=b. Nonemptiness is needed to choose that relation. The other properties follow by applying their conditions in each member.

The empty-family intersection relative to A^2 is A^2. It is reflexive, symmetric, and transitive, but if A has distinct elements a,b it contains both (a,b) and (b,a), so is not antisymmetric. The nonempty-family condition cannot be omitted. Unions need not preserve transitivity: {(1,2)} and {(2,3)} are individually transitive but their union lacks (1,3).

Partial orders

A partial order is reflexive, antisymmetric, and transitive; $(A, \preceq)$ is a poset. Two elements are comparable if $a \preceq b$ or $b \preceq a$, otherwise incomparable. A total order makes every pair comparable.

Write $a \prec b$ for $a \preceq b, a \neq b$. Element b covers a if $a \prec b$ and no c satisfies $a \prec c \prec b$. The word partial means not every pair must be comparable.

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Strict orders, restrictions, and covers

If $a \prec b \prec c$, transitivity gives $a \preceq c$. Were a=c, antisymmetry would force a=b, a contradiction. Thus the strict relation is irreflexive and transitive. Conversely, adding equality to an irreflexive transitive relation gives a partial order, since opposite comparisons would force a loop.

Restriction to $B \subseteq A$ preserves the three axioms but changes covers. Removing 2 from 1≺2≺4 makes 1–4 a cover. In a finite poset, insert intermediate elements in any noncover a≺b; finiteness prevents endless insertion, giving a chain of covers. The infinite dense order $(\mathbb{Q}, \leq)$ has no covers, so an empty cover graph cannot reconstruct it.

Positive-integer divisibility is a partial order. On nonzero integers, 1 and −1 divide each other without equality, giving only a preorder.

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Hasse diagrams

Place smaller elements lower, draw all covers, omit loops, and represent transitivity by upward paths. Only covers receive direct edges; the upward convention permits omitting arrows. If $a \preccurlyeq b$ has an intermediate c, a–b is not a Hasse edge. Drawing every pair in transitive closure adds redundant lines and hides cover structure.

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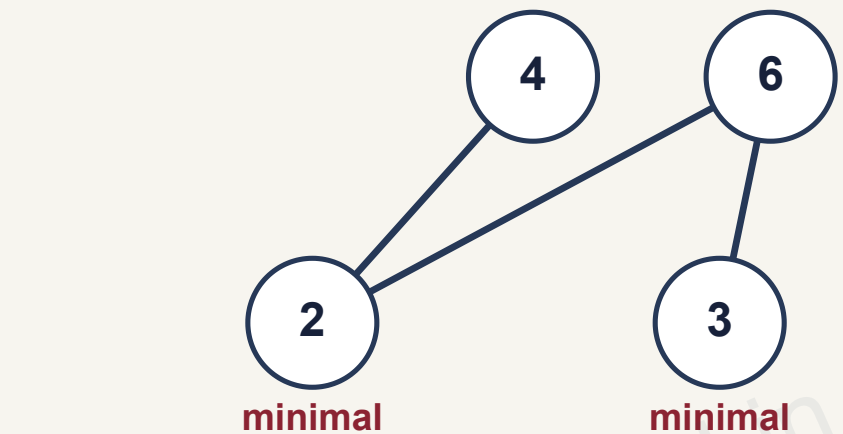
Extremal elements

Element $m \in S$ is minimal if no $x \in S$ satisfies x≺m. A maximum $M \in S$ satisfies s≼M for every s∈S. A minimum is the least element; a maximal element merely has no strictly larger element. Several minimal/maximal elements may exist, but at most one minimum/maximum.

For {2,3,4,6} under divisibility, 4,6 are incomparable and maximal, so no maximum exists. Likewise 2,3 are minimal with no minimum. Covers are exactly 2–4,2–6,3–6; neither 2–3 nor 4–6 is a cover.

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Extremal elements (continued)



cover edges

2–4 2–6 3–6

minimal: 2,3

maximal: 4,6

No minimum or maximum

Minimal means no smaller element; minimum means below every element. A maximal element does not automatically give a maximum.

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Finite extrema and a unique maximal element

Every finite nonempty poset has a maximal element. Start anywhere and move strictly upward whenever possible. No vertex repeats, since that would imply $x \prec x$; after at most $|A|-1$ moves the process stops. Descending similarly finds a minimal element.

Greatest elements are unique: candidates u,v satisfy $u \preccurlyeq v$ and $v \preccurlyeq u$, hence $u=v$. The least case is dual. If a finite poset has a unique maximal m, ascent from any x ends at a maximal, necessarily m. Thus $x \preccurlyeq m$ for every x and m is greatest.

Finiteness matters. On $\mathbb{N} \sqcup \{z\}$, give naturals their usual order and compare z only with itself. Naturals always have larger elements; z is uniquely maximal but not greatest. The empty poset has no extrema.

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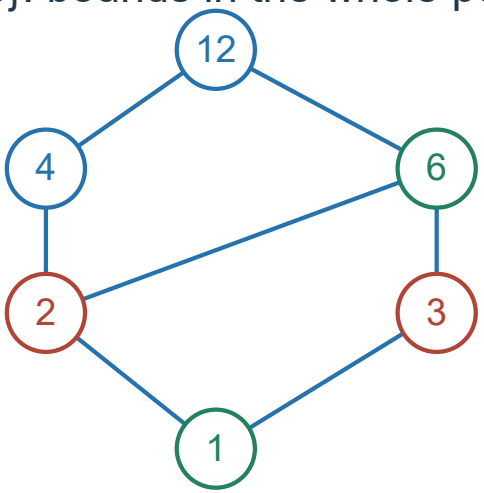
Bounds may lie outside the subset

A lower bound of $S \subseteq A$ satisfies $l \preccurlyeq s$ for all $s \in S$; an upper bound satisfies $s \preccurlyeq u$. Search in A, not necessarily S. A glb is the greatest lower bound; a lub is the least upper bound. Each is unique if it exists.

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Bounds may lie outside the subset (continued)

S={2,3}: bounds in the whole poset; glb=1, lub=6



Upward means divides; only covers are drawn.

S: red 2,3 (incomparable)

Common lower bounds: {1}; glb=1

Common upper bounds: {6,12}; lub=6

6 ≤ 12; not every upper bound is the lub.

In {1,2,3,4,6,12}, S={2,3} has lower bounds {1} and upper bounds {6,12}; glb 1 and lub 6 both lie outside S. S itself has neither minimum nor maximum. On all positive integers, meet is gcd and join lcm. Deleting elements may destroy them: {2,3,6,12} has no common lower bound for 2,3.

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A lattice's universal properties

A nonempty poset is a lattice when every pair has meet a∧b and join a∨b. Common bounds are independent of the pair's order, proving commutativity.

Common lower bounds of a,a∨b are exactly lower bounds of a: one direction requires being below a; the other uses x⩽a⩽a∨b. Hence $a \wedge (a \vee b) = a$. Reversing the order gives dual absorption. Both bracketings of a three-element meet are the glb of {a,b,c}; uniqueness proves associativity, and likewise for joins.

Repeated meet/join over a finite lattice yields bottom/top. Order reversal interchanges upper/lower, minimal/maximal, join/meet. Top and bottom alone need not give a lattice; a lattice need not be distributive. Chapter 15 adds distribution and complements, so Boolean conclusions do not apply to arbitrary posets.

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Topological sorting with proof and trace

A finite strict order is a DAG because a cycle would give $a \prec a$. Conversely, reflexive reachability in a DAG is a partial order: mutual reachability between distinct points would create a cycle.

Repeatedly output and remove a currently minimal element. Finite descent guarantees its existence. If originally $a \prec b$, b cannot become minimal before a is removed. Thus the output preserves every comparison and is a linear extension.

Divisors already output	Current minima	Smallest numerical choice
None	{1}	1
1	{2,3}	2
1,2	{3,4}	3
1,2,3	{4,6}	4
1,2,3,4	{6}	6
1,2,3,4,6	{12}	12

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Topological sorting with proof and trace (continued)

Cover adjacency lists and indegrees process initialization and each edge once in O(n+m). This excludes finding covers from a full relation matrix.

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Linear extensions and Catalan words

For distinct primes p,q, divisors of $p^{n-1}q$ form chains $A_i = p^i$, $B_i = p^i q$, 0≤i<n. Each chain increases with i; across chains, $A_i \preceq B_j$ iff $i \le j$, and no B lies below an A.

Record an extension as n A's and n B's. Within each chain, occurrence order determines indices uniquely. Legality is equivalent to every prefix having at least as many A's as B's. Before the (j+1)st B, A_j must appear; that also ensures all A_i for $i \le j$ have appeared. The reflection bijection of Chapter 1 gives

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}.$$

For n=3, the five words are AAABBB,AABABB,AABBAB,ABAABB,ABABAB. Replace successive A's by 1,2,4 and B's by 3,6,12 to recover all five extensions of the divisor poset of 12.

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Equivalence relations and classes

A reflexive, symmetric, transitive relation is an equivalence relation. Define $[a] = \{x : x \sim a\}$. Classes are equal or disjoint and partition A.

If x∈[a]∩[b], symmetry/transitivity give a~b. Any y∈[a] then satisfies y~b, proving [a]⊆[b]; reversing roles proves equality. A representative may label a class arbitrarily, but the class itself is a fixed set.

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Integers modulo five

Define a∼b iff a−b=5k for an integer k. Reflexivity uses k=0; symmetry negates k; transitivity adds two multiples of five. The classes are

$$[r] = \{5k + r : k \in \mathbb{Z}\}, \qquad r = 0, 1, 2, 3, 4.$$

They partition the integers. The quotient is $\mathbb{Z}/5\mathbb{Z}$. Chapter 14 adds ring operations, whose well-definedness must be proved before asserting that additional algebraic structure.

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Equivalence and partition are inverse constructions

Classes cover A since $a \in [a]$. If $x \in [a] \cap [b]$, $a \sim x \sim b$ gives $a \sim b$. Every $y \sim a$ then satisfies $y \sim b$, proving inclusion and, by reversal, equality. Thus $a \sim b$ iff $[a]=[b]$, and distinct classes are disjoint.

From a partition, relate elements in the same block. Each element's unique block proves reflexivity; symmetry is immediate. If $a,b \in B$ and $b,c \in C$, the common b forces $B=C$, proving transitivity. The block containing a is precisely $[a]$.

Taking classes and then the same-block relation recovers the original relation. Taking the relation and then classes recovers the original blocks. These are inverse bijections. The empty partition has no blocks, including no empty block, and corresponds to the empty relation on the empty set.

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Representative independence on quotients

For f:A→B define $a \sim_f b$ iff f(a)=f(b). Equality's properties give equivalence; projection π(a)=[a] is onto the quotient. Set $\bar{f} : A/\!\sim_f \to f(A)$, $\bar{f}([a]) = f(a)$. Changing representatives leaves the value unchanged, so it is well-defined. Equal values imply $a \sim_f b$ and equal classes, proving injection; every image has a preimage, proving surjection. Therefore

$$A/\!\sim_f \longleftrightarrow f(A), \qquad f = \bar{f} \circ \pi.$$

Generally g factors as $\bar{g} \circ \pi$ iff it is constant on every class. Equal projection values prove necessity; defining $\bar{g}([a]) = g(a)$ proves sufficiency. Surjectivity of π makes the factor unique.

Modulo five, [a]↦a mod 5 is well-defined; [a]↦a^2∈Z is not, since [1]=[6] but 1≠36. Quotient operations require the same representative check.

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Class sizes, counts, and refinement

For class sizes $s_1, \ldots, s_k$, $R = \bigcup_i (B_i \times B_i)$ is a disjoint union. Thus $|R| = \sum_i s_i^2$, with $\sum_i s_i = n$, giving $n \le |R| \le n^2$ and $|R| \equiv n \pmod 2$. Range and parity are only necessary conditions.

S(n,k) counts partitions into k unlabeled nonempty blocks. A new element either forms a singleton or joins one of k old blocks, giving

$$S(n,k) = S(n-1,k-1) + kS(n-1,k), \qquad S(0,0) = 1.$$

Out-of-range values are zero. The Bell number $\sum_k S(n,k)$ counts equivalence relations.

Write $P \preccurlyeq Q$ when every P-block lies in a Q-block. Inclusion gives reflexivity/transitivity. Under mutual refinement, $B \in P$ lies in $C \in Q$, itself in $D \in P$. Nonempty disjoint blocks force B=D, then B=C; hence P=Q. This order is equivalent to $R_P \subseteq R_Q$.

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DFA length invariant and recurrence

For a complete DFA $(Q, \Sigma, \delta, q_0, F)$, define

$$p \equiv_k q \iff \forall |w| \leq k : [\delta^*(p, w) \in F \iff \delta^*(q, w) \in F].$$

Empty input gives $P_0 = \{F, Q \setminus F\}$, omitting empty blocks. A nonempty word is av, with |v|≤k, so

$$p \equiv_{k+1} q \iff (p \equiv_0 q) \wedge \bigwedge_{a \in \Sigma} [\delta(p, a) \equiv_k \delta(q, a)].$$

Partitions only get finer. Signatures may include the current block label and each input's successor-block label; retaining the current label does not over-refine because the new relation is contained in the old.

At a fixed point $P_{k+1} = P_k$, same-block states have same-block successors under every input. Induction on suffix length proves all-word agreement: empty input agrees, and av reaches equivalent successors before applying the induction hypothesis to v. Conversely behaviorally equivalent states never split at any finite round. The stable partition is exactly behavioral equivalence.

A complete six-state refinement trace

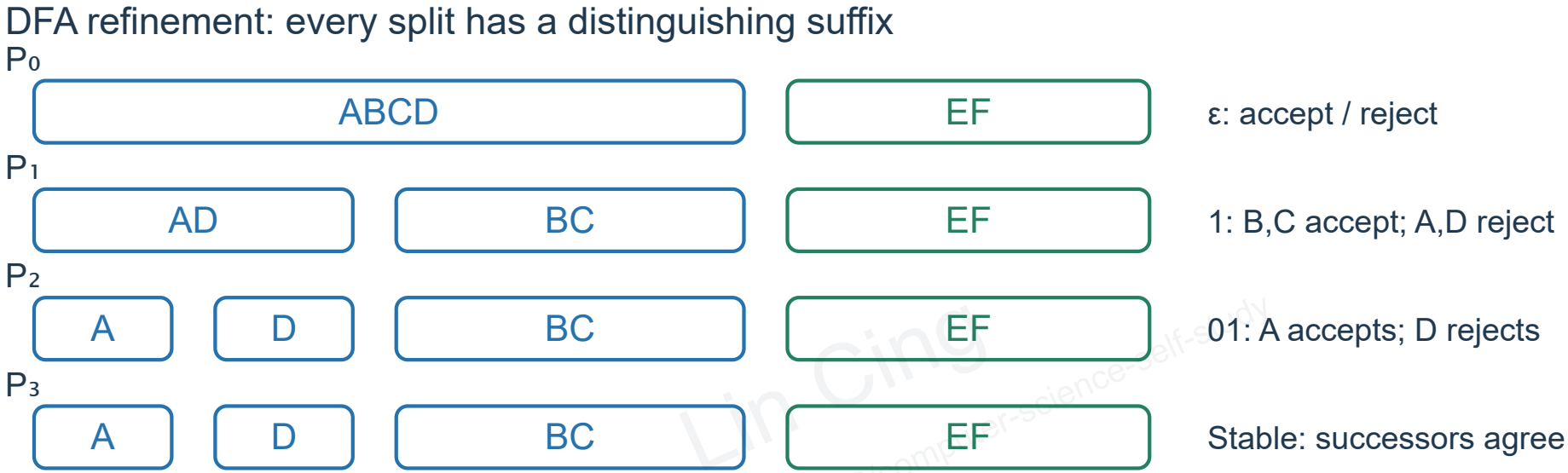
Start at A, accept {E,F}, and use alphabet {0,1}.

State	0	1	Accepts empty input?
A	B	C	No
B	D	E	No
C	D	F	No
D	D	D	No
E	E	E	Yes
F	E	F	Yes

The words ε,0,1,00,01,11 reach A,B,C,D,E,F respectively, proving all states reachable.

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A complete six-state refinement trace (continued)



P_1 splits ABCD into AD and BC according to acceptance after input 1. P_2 splits AD because input 0 reaches different P_1 blocks. P_3=P_2. Checking one edge or one state pair is insufficient.

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Quotient DFA and the lower bound

Stable blocks U={A}, V={B,C}, D={D}, T={E,F} give representative-independent $\delta'([p], a) = [\delta(p, a)]$.

Block	0	1	Accepting?
U	V	V	No
V	D	T	No
D	D	D	No
T	T	T	Yes

Empty suffix distinguishes T from each other block; 1 distinguishes U,V and V,D; 01 distinguishes U,D. If v distinguishes successors under a, av distinguishes original states. First-split rounds therefore recover witnesses backward.

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Quotient DFA and the lower bound (continued)

Choose an access prefix for each reachable class. If another same-language DFA sends prefixes x,y for two distinct classes to one state, every common suffix has the same acceptance there. A witness w makes exactly one of xw,yw accepted in the original, contradiction. Every equivalent DFA needs at least this many states; the quotient attains the bound.

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Refinement cost and unreachable states

Each genuine refinement adds a block, so at most $n - |P_0|$ occur. Signatures cost $O(n|\Sigma|)$ per round, giving $O(n^2|\Sigma|)$ with array labels and linear grouping; comparison sorting adds its own cost.

For minimization from a specified initial state, first remove unreachable states by graph search. Otherwise classes describe every state's behavior, including classes the initial state never reaches. For a partial DFA, add one common rejecting sink and complete all missing transitions first. Missing edges cannot alternate between rejection and ignoring input.

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Mealy machines have no accepting partition

A Mealy machine $(Q, \Sigma, \Gamma, \delta, \lambda)$ emits λ(q,a) on input a, then moves to δ(q,a). Its entire output satisfies

$$\lambda^*(q, \epsilon) = \epsilon, \quad \lambda^*(q, av) = \lambda(q, a)\lambda^*(\delta(q, a), v).$$

Equivalence compares whole outputs for every word and has no accepting set. Let $p \approx_k q$ mean agreement for lengths $\leq$k. P_0 is one block Q, and

$$p \approx_{k+1} q \iff \forall a : [\lambda(p, a) = \lambda(q, a) \wedge \delta(p, a) \approx_k \delta(q, a)].$$

Compare outputs for every input, then successor blocks. DFA P_0 compares empty-word acceptance; Mealy P_1 first compares one-symbol outputs. Do not mix these indices.

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Mealy merging and output witnesses

Cells give next state/output; start at p.

State	0	1
p	q/0	r/0
q	q/0	s/1
r	q/0	s/1
s	s/1	s/1

P_0={p,q,r,s}. Output rows (0,0),(0,1),(1,1) yield P_1={p}/{q,r}/{s}. Successor signatures are (qr,qr),(qr,s),(s,s), so P_2=P_1 and q,r merge.

Input 1 distinguishes p,q with outputs 0,1; input 0 distinguishes q,s with outputs 0,1. On 01, p emits 01 and s emits 11. If t shares p's row (0,0) but goes to s on input 0, input 00 distinguishes p,t with outputs 00,01.

Mealy merging and output witnesses (continued)

Induction on word length using the output recurrence proves stability correctness. For minimality, use reachable access prefixes. Distinct future-output behaviors cannot share another equivalent machine's state, since the witness suffix would then emit identical outputs. Compare future output after the prefix, without mixing in previously emitted history.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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